

Question 3  $\Rightarrow$  Class Result Processing

## \* Algorithm

(1) Start

(2) Input  $N$ 

(3) student count = 1

(4) While student count  $\leq N$ 

sum = 0

failed subject = 0

subject count = 1

    While subject count  $\leq 5$ 

Input Marks

sum = sum + Marks

        If Marks  $< 33$ 

failed subject = 1

End If

subject count = subject count + 1

END While

avg = sum / 5

Print Sum and avg

If failed subject == 1

Print "Fail - Subject Deficiency"

    Else If avg  $\geq 80$ 

Print "Distinction"

    Else If avg  $\geq 60$ 

Print "Pass"

Else

Print "Fail"

End If

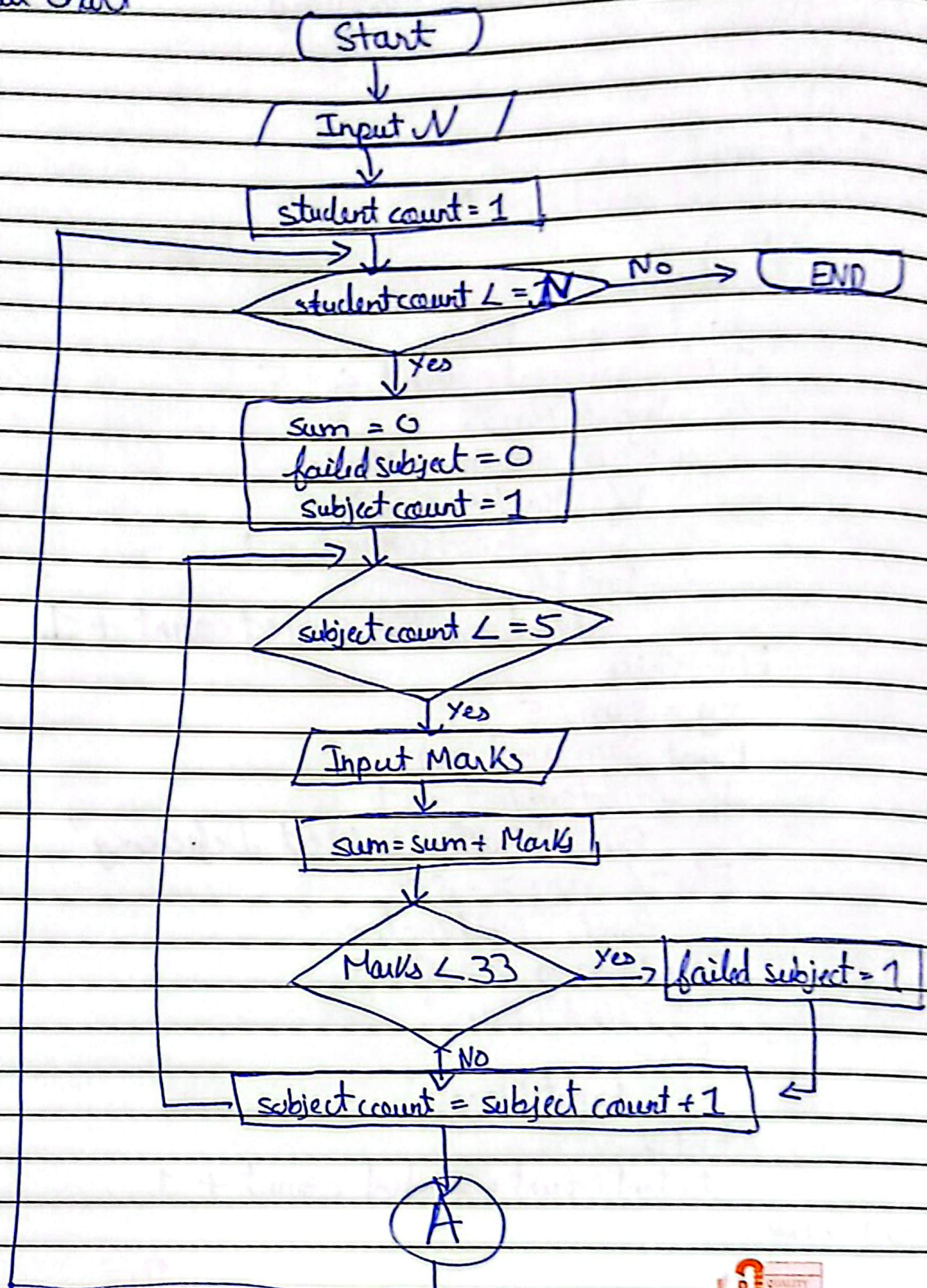
student count = student count + 1

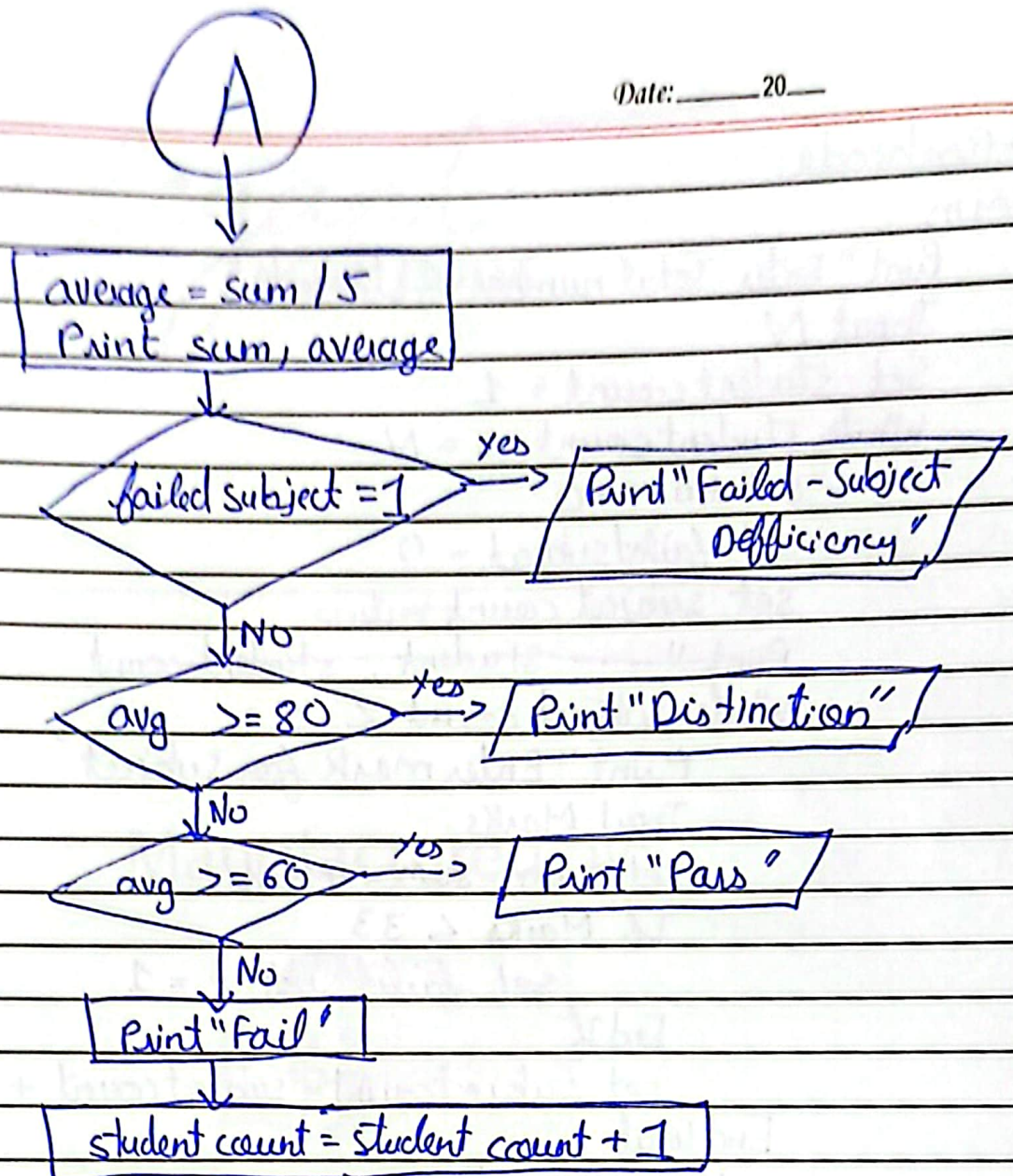
(5) End While

(6) End



## \* Flow Chart





\* Pseudocode

Begin

Print "Enter Total number of students"

Input N

Set student count = 1

While student count  $\leq$  N

set sum = 0

set failed subject = 0

set subject count = 1

Print "Student, student count"

    While subject count  $\leq$  5

Print "Enter mark for subject"

Input Marks

set sum = sum + Marks

        If Marks  $<$  33

set failed subject = 1

End If

set subject count = subject count + 1

End While

set avg = sum / 5

Print "Total marks", sum

Print "Average", avg

If failed subject == 1

Print "Fail - Subject Deficiency"

    Else If avg  $\geq$  80

Print "Distinction"

    Else If avg  $\geq$  60

Print "Fail"

End If

set student count = student count + 1

End While

End



## \* Problem Analysis Chart (PAC)

| Given Data                                                                      | Required Process                              | Output                   |
|---------------------------------------------------------------------------------|-----------------------------------------------|--------------------------|
| * Total students (N)                                                            | * Loop for N students                         | * Sum                    |
| * 5 subject marks per student                                                   | * Inner Loop for 5 subjects                   | * average                |
| * Passing mark per subject is 33                                                | * Sum the 5 marks and calculate the average   | * Classification status. |
| * $\text{average} \geq 80 \Rightarrow \text{"Distinction"}$                     | $\hookrightarrow \text{avg} = \text{sum} / 5$ |                          |
| * $\text{average} \geq 60 \text{ and } < 80$<br>$\hookrightarrow \text{"Pass"}$ | * Track if any individual mark is $< 33$      |                          |
| * $\text{average} < 60$<br>$\hookrightarrow \text{"Fail"}$                      | <del>Exit</del>                               |                          |
| * Any marks $< 33$<br>$\hookrightarrow \text{"Fail - Subject Deficiency"}$      |                                               |                          |

## \* IPO Chart

| INPUT                      | Processing                                                     | Outputs                  |
|----------------------------|----------------------------------------------------------------|--------------------------|
| * N (Numbers of students)  | (1) Input                                                      | * Sum                    |
| * marks                    | (2) Loop student count from 1 to N                             | * Average                |
|                            | (a) $\text{sum} = 0$ , failed subject = 0<br>subject count = 1 | * Classification results |
|                            | (b) Loop subject count 1 to 5                                  |                          |
|                            | * Input Marks                                                  |                          |
|                            | * $\text{Sum} = \text{Sum} + \text{Mark}_i$                    |                          |
|                            | * If Marks $< 33$                                              |                          |
|                            | failed subject = 1                                             |                          |
|                            | * subject count + 1                                            |                          |
|                            | (c) $\text{avg} = \text{sum} / 5$                              |                          |
| (d) Student count $\neq 1$ | (d) If failed subject = 1<br>Print "Fail-Subject Deficiency"   |                          |
|                            | Else If $\text{avg} \geq 80$ Print "Distinction"               |                          |
|                            | Else if $\text{avg} \geq 60$ print "Pass" Else Print "Fail"    |                          |

